

# Yangruirui ZHOU

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## EDUCATION

**Boston University, Boston, MA**

09/2020-01/2026

Doctor of Philosophy (Ph.D.) in Computer Engineering

**GPA:** 3.91/4.00

**Advisor:** Douglas Densmore

**Thesis:** AI-driven design automation for microfluidics and multicellular systems

**Research Focus:** AI-enabled microfluidic CAD, formal verification, and multicellular biological system design.

**University of Electronic Science and Technology of China (UESTC), Chengdu, China**

09/2016-06/2020

Bachelor of Engineering in Software Engineering (*Elite program*)

**GPA:** 3.94/4.00 (Rank: 6/740)

**Thesis:** Analysis of Postoperative Applications of Pose Tracking Algorithms.

## AWARDS & HONORS

iGEM "Gold medal" (Student Mentor, wiki: <https://2024.igem.wiki/bostonu/>)

11/2024

China Youth Science and Technology Innovation Award (Nominated by UESTC)

05/2020

**Most Outstanding Students of UESTC 2019** (成电杰出学生, **10/5000** in UESTC, **1/740** in department)

12/2019

iGEM "Gold medal" and "Best Software Project" (wiki: <https://2019.igem.org/Team:UESTC-Software>)

11/2019

Outstanding Graduates Award of UESTC

10/2019

"Wu Liang Ye" Enterprise Scholarship (**2/740** in Software Engineering department)

09/2019

## TECHNICAL SKILLS

**AI & Data Science:** Graph Algorithm Library (NetworkX), ML/AI/DS Library (PyTorch, TensorFlow, NumPy)

**Systems & Software:** Python, C/C++, Java, TypeScript, JavaScript, HTML, Linux Shell, LaTeX

**Biological & Microfluidic Skills:** CNC Milling, Laser Cutting, Benchmark Collection and Evaluation

**CAD Tools:** Fusion 360, Microfluidic Devices Manual Design

## RESEARCH EXPERIENCE

**Research Assistant** | Boston University, CIDAR lab

Boston, MA | 09/2020-Present

Supervisor: Prof. Douglas Densmore

- **Neptune -- Hierarchical Microfluidic Design Automation System (CUHK collaboration)**
  - Developed a full-stack microfluidic design automation system enabling end-to-end synthesis from high-level behavioral specifications to physical layout generation.
  - Developed a DSL and compiler that transform behavioral design inputs into structured connectivity graphs, which are further processed by external placement and routing algorithms; the DSL is currently supported by all five major mainstream large language models for assisted program synthesis.
  - Built a web-based interface tightly integrated into the lab's existing 3DuF design environment, enabling direct visualization, interactive editing, and manual refinement of synthesized microfluidic designs.
- **Oriole -- Genetic Circuit Partitioning for Multicellular Computation (MIT collaboration)**
  - Developed a sub-graph partitioning algorithm enabling implementation of complex computational functions (e.g., cryptographic hashing) in multicellular systems under biological and hardware constraints.
  - Designed a constraint formalization layer that encodes biological and experimental constraints into computational representations, enabling automated design and optimization in genetic circuit synthesis.
- **Vespa -- In-silico Logic-level Constraint-based Validation for Large-scale Microfluidic Systems**
  - Developed an automated in-silico validation system for microfluidic workflows, enabling simulation-based verification and reliability testing prior to wet-lab execution.
  - Constructed benchmark datasets and logic-level constraint representations for evaluating and optimizing

microfluidic design automation methods.

- **Other Works --**

- Designed and fabricated modular microfluidic systems supporting multi-modal experimental workflows for studying cell–cell communication in controlled environments.
- Built a natural language interface using large language models (Claude, GPT, Gemini, Qwen, DeepSeek) enabling users to interact with microfluidic design workflows without programming.

**iGEM Team Student Mentor** | BostonU (Wiki: <https://2024.igem.wiki/bostonu/>) Boston, MA | 05-10/2024

- Led computational and software design for iGEM 2024 project, bridging biological objectives with system-level modeling and implementation.
- Mentored students in system design and problem decomposition for biological engineering workflows.

**iGEM Team Member** | UESTC (Wiki: <https://2019.igem.org/Team:UESTC-Software>) Chengdu, China | 05-10/2019

- Built a cross-database sequence-to-protein mapping pipeline improving annotation between iGEM Registry and UniProt.
- Developed a ranking and filtering model improving alignment accuracy to >95% and increasing retrieval coverage by ~400%.

**Student Research Internship** | UCSB, Four “T”s lab Santa Barbara, CA | 06-08/2019

Supervisor: Prof. Matthew Turk and Prof. Tobies Hollerer

- Reimplemented a state-of-the-art pose tracking algorithm for local evaluation and deployment.
- Benchmarked performance against a novel method using standardized datasets.

## **PROFESSIONAL SERVICES & ACTIVITIES**

|                                                                                               |                        |
|-----------------------------------------------------------------------------------------------|------------------------|
| Mentor, <b>EBRC Mentorship for Undergraduate and Master’s Students Program (EMUMS)</b>        | 04/2025-Now            |
| Direct Mentor, <b>Eric Xie</b> , Undergraduate Research Assistant, CIDAR Lab                  | 06/2023-Now            |
| International Workshop on Bio-Design Automation (IWBD’25), <b>Oral Presentation</b>           | 09/2025                |
| International Genetically Engineered Machine competition (iGEM’19, iGEM’24)                   | 11/2019, 11/2024       |
| Inventor/Contributor, <b>Patent: Agricultural Heavy Metal Biosensor-Integrated Device</b>     | 05-11/2024             |
| Student Mentor of iGEM team: BostonU                                                          | 04-10/2024             |
| Chicago BioEngineering Conference (CBEC’24)                                                   | 09/2024                |
| Teaching Assistant, <b>EC504: Advanced Data Structures and Algorithms</b> , Boston University | 01-05/2022, 01-05/2023 |
| Engineering Biology Research Consortium (EBRC’23)                                             | 06/2023                |
| International Workshop on Bio-Design Automation (IWBD’22), Poster Presentation                | 10/2022                |

## **PUBLICATIONS**

**Google Scholar** (<https://scholar.google.com/citations?user=Zel-iSQAAAAJ&hl=en>)

### **Journals:**

- [J1, IEEE-TCAD] **Zhou, Y.**, Oliveira, S. M., Sanka, R., McIntyre, D., & Densmore, D. (2024). Vespa: Logic-Level Constraint-Based Validation for Continuous-Flow Microfluidic Devices.
- [J2, Nature Chemical Biology] Padmakumar, J. P., Sun, J. J., Cho, W., **Zhou, Y.**, Krenz, C., Han, W. Z., ... & Voigt, C. A. (2024). Partitioning of a 2-bit hash function across 66 communicating cells.
- [J3, IEEE-TCBB] **Zhou, Y.**, Voigt, C. A., & Densmore, D. (2025). Constraint-Based Sub-Graph Partitioning for Multi-Cellular Biological Networks.

### **Conferences & Workshops:**

- [C1, IWBD’22] A Conceptual Interactive Microfluidic Design and Control Workflow
- [C2, IWBD’25] LaNVis: Biological Constraint-based Large Network Visualizer (Accepted for oral presentation)

### **Manuscripts in Preparation:**

- [J4, in progress, targeting Nature Computational Science] *Neptune: Web-based Automated Design and Synthesis Platform for Microfluidic Devices.*